Xiyao Long. Time, Clocks, and the Ordering of Events in a Distributed System

- A short (3-4 sentence) summary
- They define a strict definition of total ordering of two events to ensure event a happens before event b regardless of whether event a or b are in the same processes. This requirement inspired the implementation of logical clocks. To ensure total ordering, some problems of exclusivity need to be solved. Examples of the problems include exclusive possession of a resource; resources should be granted only in the order of the requests, mechanism for ensuring all requests are granted.

- List the paper's major contributions (these may or may not match what the authors claim)
  I think the major contribution of the paper is their solution for ensuring the total ordering of requests in a process where we consider both its requests queue and also the timestamps of messages coming from all other process. This handling ensures the exclusive access of some resource and the access happens in an order defined by message time stamps.

- What's the greatest technical strength in the paper?
  I like the clear definition of "total ordering" in the paper and the discussion that in a distributed system, a failure is likely to be only identifiable in the sense of physical time and propose the notion of "total delay" and "minimum delay" of the receiving of a message.

- What are some limitations of the techniques described?
  I think the authors did not establish the connection between logical clocks and physical time clocks well enough. They use logical clocks to introduce the concept of "total ordering", but did not use the term "logical clock" anymore in the explanation of physical time defined algorithm in the second part.

- What are some ways the paper's results could be extended, improved, or better evaluated?
  This is a conceptual paper so there's no experimental result. It's nice that the authors use mathlike language to rigorously define the problems and solutions, but it will be nice if they raise several examples for the problems and solutions described to help readers better interpret their concepts.

- Does this paper suggest solutions (or approaches) to other problems?
  The author cited some other paper about handling failures from individual processes in the distributed application

- Are there parts you didn't understand?
  I'm a little confused about how the authors specialize IR1 and IR2 into IR1' and IR2'. Specifically, why it's IR1 not message specific but IR1' talked about receival of message, and why the condition is "if Pi does not receive a message"; And secondly, why it's necessary to set a minimum delay for a second process receiving some message from another process.